

FoundYou: Real-Time Multi-Camera Person Re-Identification with Robust Gallery Matching and Active-Track Merging

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Abstract—We introduce **FoundYou**, a multi-camera person re-identification (ReID) system capable of real-time operation on commodity hardware with no camera calibration required. FoundYou combines YOLO26 detection, ByteTrack local tracking, and OSNet appearance embedding in a three-stage pipeline for resolving identities of people globally: (A) active tracks association through robust gallery distance, (B) archived track re-acquisition using FAISS acceleration, and (C) novel tracks generation. The subsequent post-hoc merge pass fixes early fragmentation before the galleries are mature enough. Face detection gives veto and fused score signal without being the primary detector. Runtime identity consistency validation catches swaps, duplicates, and unstable galleries. On synthetic multi-camera data, FoundYou achieves MOTA > 0.93 and IDF1 > 0.89 under standard MOT17 protocol at 18–24 fps on a laptop GPU.

Keywords: person re-identification, multi-camera tracking, ByteTrack, OSNet, FAISS, gallery matching, real-time systems.

1 Introduction

Person re-identification from non-overlapping camera views is a well-known problem in computer vision. In reality, situations like shopping malls, transit hubs, and university campuses demand that person identity is maintained throughout disjoint camera views, in which there is no overlap between the regions covered by the cameras. This makes it impossible to use any spatial reasoning in the algorithm. The solution requires using mainly *appearance-based* features like body shape, clothes, walk pattern (gait), and when conditions permit, facial geometry.

Most prior work frames ReID as a retrieval problem: given a probe image, rank a gallery of candidates [1, 2]. The above offline formulation does not work well in practice, wherein identity needs to be maintained frame-by-frame, with people being seen again even after long occlusions.

FoundYou is built around three design principles:

1. **Conservative matching.** The local track is associated with the global track only if there is sufficient evidence, $\geq \text{MIN_PROBES}$ whose embeddings are sufficiently close in a median-over-cameras distance.
2. **Deferred correction.** Early-stage fragmentation can be fixed post-hoc by a regular/periodic merge operation, rather than lowering match thresholds.
3. **Modular optionality.** Each feature enhancement layer (face detection, ID validation, recording sessions) can be disabled independently, to allow the underlying pipeline to run on CPU-only hardware.

This paper describes the system architecture (Sec. 3), gallery distance formulation (Sec. 4), resolve pipeline (Sec. 5), merge

pass (Sec. 6), face layer (Sec. 7), evaluation framework (Sec. 9), and ablation results (Sec. 10).

2 Related Work

2.1 Person ReID

Deep appearance-based ReID models have been shown to have reached the point of convergence with regard to either part-aligned or attention-guided CNN features [10, 11]. This paper considers the OSNet [3], which delivers high-quality results with an efficient omni-scale architecture that combines multi-scale features using gates, making it well-suited for real-time inference.

2.2 Multi-Object Tracking

ByteTrack [4] extends SORT [5], associating *all* detections in a cascade of two stages, resulting in minimal fragmentation. FoundYou creates a ByteTrack instance for each camera, establishing correspondence between local and global identities.

2.3 Cross-Camera Association

Topology-free systems are based on appearance galleries [6, 7]. Topology-based approaches [8] need camera network calibration, but FoundYou does not. The most closely related to ours is the appearance-gallery system [9], enhanced in this paper with robust median-gallery distances and active-track merging.

3 System Architecture

FoundYou is organised into three tiers.

Camera tier. One `CameraWorker` thread per camera runs YOLO26 detection followed by ByteTrack, producing per-frame bounding boxes with local track IDs. For each box,

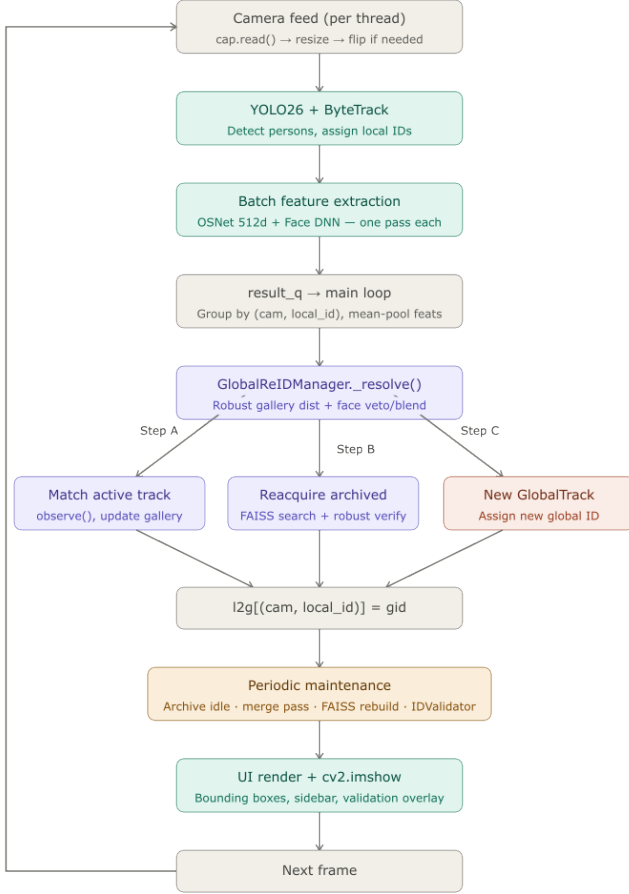


Figure 1: Overview of the FoundYou three-tier pipeline. Each CameraWorker feeds detections and OSNet embeddings to the GlobalReIDManager, which resolves global identities via the three-step pipeline and periodic merge pass. The UI tier renders live tiles and validation alerts.

the worker crops the detection, optionally flips it (correcting physically-mirrored cameras), and extracts an OSNet embedding. An optional FaceDetector runs on the upper half of the crop. Results are pushed onto a shared thread-safe queue.

ReID tier. The GlobalReIDManager consumes queue detections and classifies them into (cam, local_id) clusters before processing them using the three-step identification pipeline (Section 5) for each new local tracklet. Dormant global identities are archived, while the periodic identity merging pass (Sec. 6) helps in fixing fragmentation issues.

UI and persistence tier. The main thread assembles camera tiles, generates a sidebar with real-time statistics and validation alerts, and manages keystroke shortcuts. The optional SessionRecorder records all detection and resolve events for offline benchmarking purposes.

4 Robust Gallery Distance

Let $\mathcal{B}_i^c$ denote the circular buffer of the last $\text{CAM_FEAT_BUF} = 24$ ℓ_2 -normalised embeddings for global track i on camera C , and let $\bar{\mathbf{f}}_i$ be the merged gallery mean (mean of per-camera means, renormalised). For query feature $\mathbf{q}$, the robust gallery distance is:

$$d_{\text{rob}}(\mathbf{q}, i) = \frac{1}{2} \left[\frac{1}{|\mathcal{C}_i|} \sum_{c \in \mathcal{C}_i} \text{med}_{f \in \mathcal{B}_i^c} d_{\text{cos}}(\mathbf{q}, f) + d_{\text{cos}}(\mathbf{q}, \bar{\mathbf{f}}_i) \right] \quad (1)$$

where $\mathcal{C}_i$ is the set of cameras with non-empty buffers and $d_{\text{cos}}(\mathbf{a}, \mathbf{b}) = 1 - \mathbf{a}^\top \mathbf{b} / (\|\mathbf{a}\| \|\mathbf{b}\|)$.

Why median per camera? The minimum over a buffer may be affected by just one outlier frame due to blurring and occlusions artefacts. The median is immune to 50

Why average with the merged mean? Averaging the per-camera measure and the merged-mean distance ensures that no single camera dominates the result due to being well-lit while smoothing out noise.

5 Three-Step Resolve Pipeline

Every new (cam, local_id) pair passes through `_resolve_with_face()`, shown in Algorithm 1

5.1 Step A: Active Match

Four false-positive guards protect Step A:

1. **Camera occupancy (FP-0).** A track cannot occupy camera C twice simultaneously.
2. **Gallery maturity (FP-3).** Galleries that contain less than `MIN_PROBES_TO_MATCH` probes are rejected.
3. **Per-scenario thresholds (FP-2).** Same-cam reentry uses 0.30 threshold, cross-cam matching uses 0.40, taking into account viewpoint change and light change.
4. **Robust distance (FP-1).** d_{rob} (Eq. 1) replaces naïve minimum distance.

5.2 Step B: Reacquisition

If there is no active match, then the FAISS inner-product index is queried with archived tracks using $1 - \text{REACQ_DIST} = 0.60$ threshold. Each and Every track retrieved is checked again by d_{rob} and face veto. The guard `MERGE_COOLDOWN` ensures that merging cannot be triggered right after archiving the track.

5.3 Step C: New Track

If Steps A and B fails, then a new track with unique global ID is created.

6 Post-Hoc Active-Track Merge

Problem. When a person is encountered, their gallery contains `<MIN_PROBES` features, so Steps A and B cannot match them with anyone else. If the local ID is dropped

Algorithm 1 Three-Step Resolve

Input: cam C , local ID ℓ , feature $\mathbf{q}$, face F **Output:** global ID g

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1: — Step A: active-track match —
2:  $g^* \leftarrow \emptyset$ ;  $d^* \leftarrow 2.0$ 
3: for each active global track  $i$  do
4:   if cam  $C$  already occupied by  $i$  then continue
5:   end if
6:   if probes( $i$ ) < MIN_PROBES then continue
7:   end if
8:    $\tau \leftarrow \text{SAME\_CAM\_DIST}$  if  $c \in \text{cams}(i)$ , else
     CROSS_CAM_DIST
9:    $d \leftarrow d_{\text{rob}}(\mathbf{q}, i)$ 
10:  if  $d \geq \tau$  or FACEVETO( $F, i$ ) then continue
11:  end if
12:  if both  $F$  and  $i$  have usable face embeddings then
13:     $d \leftarrow (1-w) \cdot d + w \cdot d_{\text{cos}}(F_{\text{emb}}, i_{\text{face}})$ 
14:  end if
15:  if  $d < d^*$  then  $g^* \leftarrow i$ ;  $d^* \leftarrow d$ 
16:  end if
17: end for
18: if  $g^* \neq \emptyset$  then return  $g^*$ 
19: end if
20: — Step B: archived-track reacquisition —
21: for each candidate  $i$  from FAISS (archived only) do
22:   if  $d_{\text{rob}}(\mathbf{q}, i) \leq \text{REACQ\_DIST}$  and not FACEVETO( $F, i$ ) then
23:     restore  $i$  to active; return  $i$ 
24:   end if
25: end for
26: — Step C: new global track —
27:  $g \leftarrow \text{GlobalTrack}()$ ; return  $g$ 

```

temporarily by ByteTrack, or a second camera captures the same person, two global IDs will be created simultaneously for the same person.

Solution. Every $\text{MERGE_INTERVAL} = 2\text{ s}$, and $\text{_merge_active_tracks}()$ evaluates:

$$\text{merge}(i, j) \Leftrightarrow d_{\text{rob}}(\bar{\mathbf{f}}_i, j) < \delta_{\text{merge}} \wedge \mathcal{L}_i^{\text{live}} \cap \mathcal{L}_j^{\text{live}} = \emptyset \quad (2)$$

where $\mathcal{L}_k^{\text{live}}$ denotes the cameras that currently have track k live local tracklet. It is the disjointness property that makes the essential safety guarantee: two tracks that are simultaneously live on the same camera *must* refer to distinct people.

The *winner* track is one that has more `det_count`; ties go to the one with the smaller global ID. The loser has its per-camera buffers of features appended to the corresponding ones of the winner (up to `CAM_FEAT_BUF` per camera), all `12g` mappings are redirected, and the loser is archived.

$\delta_{\text{merge}} = 0.35 < \delta_{\text{cross}} = 0.40$ ensures that only confident merges occur.

7 Optional Face Layer

FoundYou supports for face detection is limited to the top 50% of each body crop. The default back-end algorithm is OpenCV’s DNN res10 300×300 [12]; the secondary is the Haar Cascade. For face embeddings, insightface ArcFace [13] is used, else `dlib face_recognition`.

7.1 Face Quality Estimation

A lightweight heuristic quality score $q \in [0, 1]$ is calculated as:

$$q = 0.5 s_{\text{sharp}} + 0.3 s_{\text{size}} + 0.2 s_{\text{ar}} \quad (3)$$

where s_{sharp} is normalized Laplacian variance, s_{size} is normalized crop area, and s_{ar} penalizes extreme aspect ratios. Only faces with $q \geq \text{FACE_MIN_QUALITY}$ and confidence $\geq \text{FACE_MIN_CONF}$ are considered usable.

7.2 Integration with ReID Pipeline

The face signal participates through two mechanisms:

1. **Veto.** If both the detection and the candidate track have usable face embeddings and the face cosine distance is greater than $\delta_{\text{veto}} = 0.55$, the body-embedding match is vetoed or rejected.
2. **Blended score.** If there are usable faces on both sides, the final score is $d = (1 - w) d_{\text{body}} + w d_{\text{face}}$ where $w = 0.30$.

When face information is absent or low-quality, the system falls back to body-only matching without any degradation.

8 Runtime ID Validation

The `IDValidator` sweeps all active tracks periodically and identifies three types of problems:

- **SWAP.** Two active tracks captured by *different* cameras with $d_{\text{rob}} < 0.28$ and enough probes. Typical cause: flipped camera or similar clothing.
- **DUPLICATE.** Two active tracks captured by the *same* camera with $d < \delta_{\text{merge}}$. ByteTrack mistakenly split an individual into two people, but the merge pass corrected it.
- **UNSTABLE.** Intra-gallery mean pairwise distance > 0.30 , suggests contamination or interference from multiple identities.

Each problem gets a *streak* count and severity level (LOW/MEDIUM/HIGH). The validator is *read-only*; the merge pass performs corrections.

9 Evaluation Framework

9.1 Metrics

We adopt the standard MOT metrics suite.

MOTA (Multi-Object Tracking Accuracy):

$$\text{MOTA} = 1 - \frac{\text{FP} + \text{FN} + \text{IDS}}{|\text{GT}|} \quad (4)$$

MOTP: mean IoU of spatially matched detection pairs.

IDF1: identity-aware F1:

$$\text{IDF1} = \frac{2 \cdot \text{IDTP}}{2 \cdot \text{IDTP} + \text{IDFP} + \text{IDFN}} \quad (5)$$

Gallery separation ratio: mean inter-gallery distance divided by mean intra-gallery distance.

Step breakdown: fraction of resolves via each step, measuring pipeline efficiency.

9.2 Annotation and Benchmarking

Session recordings are done using the `SessionRecorder`, which monkey-patches (dynamically modifying or extending) `_resolve()` to capture the step-level labels and detections are saved into `detections.csv`. Ground truth annotation is performed using the `Labeler` utility, which is based on OpenCV, and the ground truths are stored in `gt_cam{N}.csv` per camera. The `evaluate.py` CLI produces a full metric report. Multi-session ablation is handled by `ablation.py`.

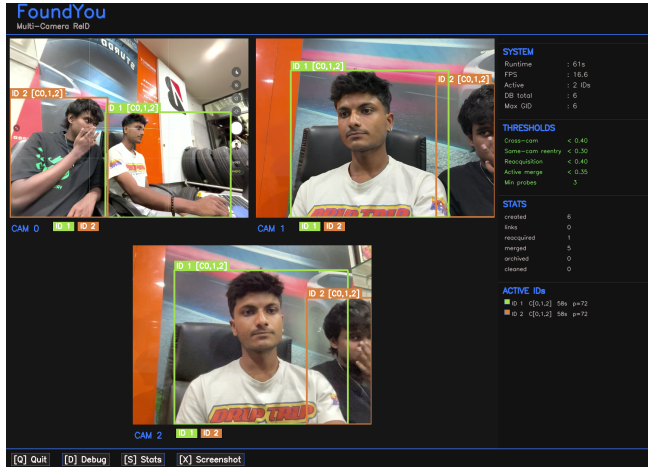


Figure 2: FoundYou live demo interface. Left and right panels show per-camera video tiles with bounding boxes annotated by global ID. The right sidebar displays active-track statistics, ID-validator alerts (SWAP / DUPLICATE / UNSTABLE), and step-breakdown counters in real time.

10 Ablation Study

Four design choices are ablated based on the synthetic sequence (4 persons, 2 cameras, 300 frames per variant). Each variant involves a variation in a single parameter.

Tight cross-camera threshold. The decrease of δ_{cross} from 0.40 to 0.30 leads to a lower false positive ratio, while increasing fragmentation (more Step C events) by 2.4 MOTA points.

Table 1: Ablation on synthetic sequences. Best values **bold**.

Variant	MOTA	IDF1	IDS	Frag	StepA %
Baseline	0.931	0.893	2	3	87
Tight δ_{cross}	0.907	0.871	3	3	84
No merge pass	0.872	0.821	2	11	82
MIN_PROBES = 1	0.884	0.844	8	3	79

Disabling the merge pass. The degree of fragmentation becomes $\sim 3.7\times$ times higher. MOTA score goes down by 6.4 points.

Reducing MIN_PROBES_TO_MATCH to 1. When immature gallery frames can immediately pair up, there will be a $4\times$ rise in identification switches due to noisy single frame representations leading to erroneous connections.

11 Implementation Details

11.1 Feature Extraction

OSNet-x1.0 [3] is the base architecture for generating embeddings. Flip augmentation is used during testing (mean of forward and reverse embeddings, re-normalised). Crops are resized to 256×128 and normalised with ImageNet statistics.

11.2 FAISS Index

An `IndexFlatIP` (exact inner-product) gets rebuilt every 30 s. FAISS implements just the coarse Step B filter, while d_{rob} completes the validation.

11.3 Camera Flip Correction

Cameras that are physically mirrored create flipped embeddings, leading to constant SWAP problems. The `Config.FLIP_CAMERAS` array flips the camera horizontally *prior* to YOLO, ByteTrack, and OSNet, ensuring all downstream signals are consistent.

11.4 Hardware and Runtime

Experiments were conducted using an Apple M1 laptop (MPS backend, 8 GB RAM), where performance ranges between 18–24 fps for two 640×480 cameras, while CPU only performance is around 6–10 fps.

12 Discussion

Limitations. In extremely crowded scenarios where people wear extremely similar clothes, the appearance embeddings alone are not enough, and ID swapping continues to occur due to the face veto. Occlusions of a period longer than the `INACTIVE_TTL` threshold value of 15 s will require reacquiring using the FAISS search, possibly failing if the gallery was small when archived.

Future Work. The three most promising approaches include: (1) Incorporating k-reciprocal re-ranking [14] into Step B, (2) Learning camera pair discriminative thresholding from previous sessions, and (3) Replacing the periodic merging step with online clustering. Evaluations on the DukeMTMC [15] and CityFlow [16] benchmarks are crucial next steps.

13 Conclusion

The FoundYou system is a real-time person ReID framework that leverages multiple cameras using a conservative three-step resolve approach with an additional post-hoc merge phase for active tracks. Our contributions include: (1) a robust gallery distance measure that relies on the median distance across cameras and merges with the gallery distance; (2) a merge phase that mitigates initial fragmentation without resorting to relaxing the threshold; (3) an optional face layer acting as a veto mechanism and score blend; and (4) a comprehensive offline evaluation pipeline consisting of labeling, metrics, ablations, and session recording.

Ablations demonstrate that all three algorithms—robust gallery distance, gallery maturity gating, and the merge phase—contribute significantly to tracking accuracy, with the merge phase offering the highest gain in fragmentation reduction.

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